1.8 Solving Absolute Value Equations

Name _____ Period ____ Date _____

Algebra I 1.8 (30-minute plan) · Student sheet

I can…

- solve absolute value equations.
- solve equations with two absolute values and check for extraneous solutions.
- write absolute value equations for real-life situations.

Vocabulary · fill in the missing word

Absolute value — |a| is the _____ from a to 0 on a number line

Absolute value equation — an equation with a _____ inside absolute value bars

Extraneous solution — a value found while solving that does not satisfy the _____ equation

Part 1 · Notes and examples (fill in each note, then work the example)

Absolute value equations: |ax + b| = c means ax + b = _____ or ax + b = _____.

No solution: |x| = −3 has _____, because absolute value is never _____.

Example 1

Solve each equation.

a. |x| = 6 b. |x + 3| = 10

Isolate first: Isolate the _____ first.

Example 2

Solve 3|2x − 1| + 4 = 19.

Check: Always check for _____ solutions.

Example 3

Solve |x − 4| = |2x + 1|.

Part 2 · Partner practice

Partner practice

Solve $|x - 2| - 3 = 5$.